CDR Course – PDD and Methodology for GHG Mitigation

CONTENTS

1 PROJECT DETAILS

1.1 Summary Description of the Project

The proposed project is biochar production facility in the Bikasandra Village of KR Pet Taluka in Mandya district of Karnataka State India which is most agricultural prosperous district in Karnataka intends to address the issue of crop residue management from the Coconuts fronds, Paddy straw, Sugarcane trash, and other waste agriculture residues and converting them to Biochar in the region while addressing the soil carbon improvement with the application of biochar

Presently agricultural residues are often burned in fields causing air pollution or left to decay in the fields with loss of potential value. Converting residues to biochar provides an economic product and climate benefits (carbon sequestration) and enriches the soil carbon, fertility and efficient water management.

Biochar is defined as carbon-rich material produced by partial oxidation (pyrolysis at ≤700 °C in the absence or limited supply of oxygen) of carbonaceous organic sources such as wood and plants, excluding fossil fuel products. The International Biochar Initiative (IBI) rules currently govern the most standardized definition of biochar, which specifies that biochar is a solid substance generated through biomass pyrolysis in an oxygen-limited environment

Objective is to produce high-quality biochar from waste agri residue and use it for soil amendment, carbon credits, meeting SD goals, CO_2 direct removal, and industrial uses while ensuring environmental compliance and worker safety, which other wise would have been burnt in the field or left to decay impacting the environment.

- The proposed project activity uses state of the art High technology continuous type Pyrolysis units combined with preprocessing units and air emission controls. The project targets coconut fronds, cane trash, paddy straw, agricultural residues.
- The project location is in Bikasandra Village (Latitude: 12.7896° N (12.789604) Longitude: 76.5710° E (76.571013)) in KR Pet Taluka, Mandya District, Karnataka State, INDIA.
- This proposed project activity is a greenfield project. As per the provision of the applied methodology VM0044, in high technology settings the total organic carbon content of the produced biochar is the basis of the GHG calculations. The value is derived from the mass of biochar, its respective organic carbon content, and the decay rate of organic carbon in the biochar taken over a period of 100 years (100-year permanence value).
- The scenario prior to the project activity is that the agricultural waste biomass (i.e. coconut fronds, cane trash, paddy straw and waste agri residues) is either left to decay or combusted. However, a survey related to the baseline scenario has been conducted in the project region and it has been established that the waste biomass (i.e. coconut fronds, cane trash, paddy

straw, and vegetable residues) only have the fate as: combustion of biomass and such burning mostly open burning which is a prevailing practice and a concern in the region

- Primary feedstocks are coconut fronds, cane trash, paddy straw, waste agri residues - Estimated total biomass collection target: >60,000 TPA (to produce ~20,000 TPA biochar, accounting for moisture, ash, and process yields).The proposed project activity is expected to generate an annual average of 40,000 tCO_2e emission reductions which is estimated based on a projected generation of biochar from Coconut frond, cane trash, paddy straw, waste agri residues etc as feedstock during all round season in each year

1.2 Audit History

For projects undergoing crediting period renewal, include the audit history of the project using the table below. For the project validation, state the validation date in the Period column. This table should include all monitoring periods, including the period of this report.

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	*N.A*	*VCS*	*Validation/verification body name*	*One year*
	…			

1.3 Sectoral Scope and Project Type

Complete the table below with information relevant for non-AFOLU projects:

Sectoral scope	13
Project activity type	Biochar production and application

Complete the table below with information relevant for AFOLU projects:

Sectoral scope	
AFOLU project category	
Project activity type	

1.4 Project Eligibility

1.4.1 General eligibility

S N	Condition	Justification
	The project activity must install and operate a new (greenfield) biochar production facility(ies) where the project proponent must 1) source waste biomass 2) produce biochar and	Eligible: The project activity is a greenfield biochar production facility. 1) The waste biomass is coconut fronds, cane trash, paddy straw, crops residues which is sourced from the local farmers 2) The proposed plant will produce biochar

	3) ensure the biochar is utilized in soil or non-soil application. GHG benefits are credited only for the biochar that is utilized in the eligible soil and non-soil applications.	3) The produced biochar will be utilized for soil application and non-soil applications only
2	Technological Scope: (i) The methodology is applicable when biochar is produced from eligible waste biomass through a thermochemical process such as pyrolysis, gasification, and biomass boilers and the biochar is subsequently applied to an end-use (soil or non-soil applications). Torrefaction and hydrothermal carbonization as processes of biochar production are excluded from this methodology. (ii) The methodology is applicable to projects using either low or high technology production facilities to produce biochar, as per the definitions of each provided in Section 3 of this methodology. (iii) The biochar producers must have a health and safety program to protect workers from airborne pollutants and other hazards.	Eligible: (i) The project activity involves pyrolysis process and subsequently applied for soil and non-soil application only. (ii) The project activity involves high technology production facilities (i.e. Pyrolysis technology with heat recovery and emission pollution control) to produce biochar. (iii) There is a proper health & safety procedure proposed under this project to support the workers with safe working environment.

3	Feed stocks and production scope: The feedstock used to produce biochar must meet all following conditions to be eligible: **a.** Feedstock must be purely biogenic waste biomass and not purpose-grown, **b.** Feedstock must have been otherwise left to decay or combusted for purposes other than energy production. Additional guidance on how to demonstrate fate of waste biomass in the absence of the project activity is provided in Appendix 2,	Eligible: **a.** The project activity involves only coconut fronds, cane trash, paddy straw, agricultural residues as feedstock, which is a biogenic waste biomass and hence not purpose-grown. **b.** In absence of the project, the feedstock would have been combusted without any energy production or similar purposes. The fate of the biomass feedstock is referred in line with the Appendix 2 of the methodology; a detailed baseline study has been conducted for the same.
	c. Feedstock must not have been imported from other countries. **d.** Feedstock must meet the sustainability conditions provided in Table 1 of the methodology.	**c.** The project boundary is limited to the Mandya district of Karnataka state in India. The feedstock is also sourced within the same region. Hence, there is no import of feedstock. **d.** The feedstock being used under the project activity is coconut fronds, cane trash, paddy straw agricultural residues which falls under the “Agriculture waste biomass”; the project is designed to use coconut fronds, paddy straw, agricultural residue directly from fields and

		processed in a centralized biomass processing operation.
4	Biochar made from a single or mixed eligible feedstock types must comply with the latest version of the IBI Biochar Testing Guidelines or the EBC Production Guidelines.	Eligible: The biochar under this project activity is produced from a mixed feedstock type, i.e. coconut fronds, cane trash, paddy straw and agricultural residues and follows the requirements of IBI Biochar guideline. The details will be provided during the course of validation & verification.
5	The waste biomass used as feedstock to produce biochar and the resulting biochar to be utilized in soil or non-soil applications may be transported via ships, boats, and vehicles other than road transportation up to a distance of 200 km. However, it must only be transported by vehicles (i.e., road transportation) for distances more than 200 km as defined under CDM Tool 12: Project and leakage emissions from transportation of freight.	Eligible: The biochar under this project activity is produced using waste biomass and shall be utilized only for soil applications in and around the fields where the biomass waste is generated. Hence there is no involvement of dedicated transportation facilities; hence Tool 12 is not applied.
6	Mineral additives such as lime, rock minerals, and ash may comprise up to 10 percent of the mass when added. If the addition exceeds 10 percent on a dry weight basis, the biochar producer must present laboratory tests indicating that the final product meets IBI Biochar Testing Guidelines or EBC Production Guidelines thresholds for organic and inorganic contaminants.	Not applicable. The production process of biochar under the project activity does not include any additive

7	Biochar is eligible to be utilized and accounted for under the methodology if it is being utilized within one year of its production. Biochar is subject to natural decay and the permanence of biochar is calculated for a period of 100 years. To adhere to the decay factor established for 100 years and prevent any decay before application, biochar must be utilized in soil or non-soil applications, as appropriate, within the first year of its production.	Eligible: The biochar produced under the project activity shall be utilized for soil and non-soil applications within one year of its production which will be tracked by digital MRV application
8	Biochar is eligible to be used as a soil amendment on land other than wetlands. Eligible land types include cropland, grassland, and forest. Biochar is eligible to be applied either to the soil surface or subsurface. For surface application, the biochar must be mixed with other substrates such as compost, manure or digestate from anaerobic digestion. For subsurface application, the biochar may be applied either as a unique soil amendment or mixed with other substrates. For any soil application, the biochar must comply with biochar material standards to avoid the risk of transferring unwanted heavy metals and organic contaminants to soil. Project proponents must meet the IBI Biochar Testing Guidelines or EBC Production Guidelines or relevant national regulations for avoiding soil contamination. Biochar shall have a hydrogen to organic carbon molar ratio (H:Corg) of less than or equal to 0.7.	Eligible: The biochar produced under the project activity shall be used for soil applications. Depending on the farmers requirement it can be used as soil amendment on land both to the soil surface and/or subsurface. In all cases, the biochar application shall meet the requirements of IBI Biochar Testing Guidelines. The details will be provided during the course of validation & verification. Eligible H: C ratio shall be <0.7

9	Biochar is eligible to be used in non-soil applications including but not limited to cement, asphalt, and any other applications where long-term storage of the biochar is possible. Only biochar produced in high technology production facilities, as defined under the methodology, is eligible to be used in non-soil applications.	Eligible The proposed plant is a high technology biochar production facility, any biochar in excess of soil amendment requirement on land will be used in non-soil applications as in steel manufacturing industries
10	Project proponents must demonstrate that biochar and/or final products are long-lived via credible evidence such as laboratory results, peer reviewed research papers or any other third party-evaluated product assessment, such as decay rate analysis, as applicable. The information provided must include the lifetime of the product in which biochar is stored long term. The resultant product must be compliant with national/international product quality standards/specifications as applicable (e.g., the American Concrete Institute Standards in the US).	Eligible Based on the biochar plant manufacturers test certificate & lab results we will demonstrate that produced in the proposed plant will be long-lived

1.4.2 AFOLU project eligibility

For AFOLU projects, describe and justify how the project is eligible to participate in the VCS Program. The response should:

- *Justify and demonstrate that all selected AFOLU project categories are appropriate and that all related category requirements are met.*
- *Provide evidence that native ecosystems have not been converted, cleared, drained, or degraded to generate GHG credits in Section 2.4.3 below.*
- *For ARR, ALM, WRC, or ACoGS project areas, provide evidence that clearing or conversion did not take place within 10 years of the project start date in Section 2.4.3 below.*

This is Non AFOLU project

1.4.3 Transfer project eligibility

For transfer projects and CPAs seeking registration, justify how eligibility conditions have been met. The response should justify how the criteria in Appendix 2 and Section 3.23 (Double Counting and Participation under Other GHG Programs) of the VCS Standard have been met.

Not applicable

1.5 Project Design

Indicate if the project has been designed as:

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

For grouped projects, provide additional information relevant to the design of the grouped project, including any eligibility criteria that new project instances must meet upon their inclusion, subsequent to the initial validation of the project.

Not applicable

1.6 Project Proponent

Provide contact information for the project proponent(s). Copy and paste the table as needed.

Organization name	
Contact person	
Title	
Address	
Telephone	
Email	*The email address domain must match that of the organization.*

1.7 Other Entities Involved in the Project

Provide contact information and roles/responsibilities for any other entities involved in the development of the project. Copy and paste the table as needed.

Organization name	
Role in the project	

Contact person	
Title	
Address	
Telephone	
Email	*The email address domain must match that of the organization.*

1.8 Ownership

The project ownership is with Shree Krishna CDR — a Special Purpose Vehicle to be incorporated as a wholly-owned subsidiary of CVC Group under the Companies Act, 2013. Incorporation to be completed prior to the validation opening meeting. CVC Group, Bengaluru, Karnataka, serves as authorized representative at pipeline listing stage. in accordance with the VCS Program specifications on project ownership, the project ownership is meant to be arising by virtue of a statutory, property or contractual right in the biochar production plants/facilities, equipment or process etc. that generates GHG emission reductions and/or removals.

1.9 Project Start Date

Project start date	*01-September-2026*
Justification	*Justify how the project start date conforms with the VCS Program requirements* The start date of the project is defined as the first instance of biochar production in a new (greenfield) facility.

1.10 Project Crediting Period

Crediting period	☐ *Seven years, twice renewable* ☒ *Ten years, fixed* ☐ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*
Start and end date of first or fixed crediting period	*01-April-2027* to *31-March-2037*

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

Indicate the estimated annual GHG emission reductions/removals (ERRs) of the project:

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Complete the table below for the first (if renewable) or fixed crediting period:

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
01-January-2026 to 31-December-2026	0
01-January-2027 to 31-December-2027	10,000
01-January-2028 to 31-December-2028	20,000
01-January-2029 to 31-December-2029	30,000
01-January-2030 to 31-December-2030	30,000
01-January-2031 to 31-December-2031	30,000
01-January-2032 to 31-December-2032	30,000
01-January-2033 to 31-December-2033	30,000
01-January-2034 to 31-December-2034	30,000
01-January-2035 to 31-December-2035	30,000
01-January-2036 to 31-December-2036	30,000
Total estimated ERRs during the first or fixed crediting period	2,31,524 (Modules 1–3; see Section 4.4 for revised calculation)
Total number of years	10
Average annual ERRs	30,000

1.12 Description of the Project Activity

Describe the project activity or activities (including the technologies or measures employed) and how it/they will achieve the GHG emission reductions or carbon dioxide removals. Describe the implementation schedule of project activity or activities.

For non-AFOLU projects:

- *Include a list and the arrangement of the main manufacturing/production technologies, systems and equipment involved. Include in the description information about the age and average lifetime of the equipment based on manufacturer's specifications and industry standards, and existing and forecast installed capacities, load factors and efficiencies.*

- *Include the types and levels of services (normally in terms of mass or energy flows) provided by the systems and equipment that are being modified and/or installed and their relation, if any, to other manufacturing/production equipment and systems outside the project boundary. Clearly explain how the same types and levels of services provided by the project would have been provided in the baseline scenario.*
- *Where appropriate, provide a list of facilities, systems, and equipment in operation under the existing scenario prior to the implementation of the project.*

For AFOLU projects:

- *For all measures listed, include information on any conservation, management or planting activities, including a description of how the various organizations, communities and other entities are involved.*
- *In the description of the project activity, state if the project is located within a jurisdiction covered by a jurisdictional REDD+ program.*

The project activity is a non-AFOLU project under VCS, mainly engaged in the process of Biochar production using high technology facilities with the coconut fronds, cane trash, paddy straw, agricultural residue as sustainable biomass waste as feedstock.

The design of the project activity includes Pyrolysis unit in a specified location for biochar production facility ensuring a practical and sustainable way of producing biochar in the project region. The technology and prescribed methods is detailed as follows:

A) BIOCHAR PRODUCTION CAPACITY:

It is proposed to set up initial capacity of one modular plant comprising 2 Nos 1 TPH Biochar units, and scale up in every 6 months. Hence, between 2026-27 and 2028-29, build up capacity to 6 x 2 Nos 1 TPH plants.

Thus, build up capacity to 30,000 TPA Biochar, out of which 20,000 tons will be sequestered as CDR project Carbon credits for CO_2 removal will be contracted @ 10,000 tons Biochar for FY 2027-28- and 20,000-tons Biochar from FY 2028-29 onwards, which can be ensured due to higher production capacity being created.

B) FEEDSTOCK SUPPLY CHAIN:

For each modular Biochar plant of 5,000 TPA (1,00,000 no’s 50 Kg bags), about 16,000 TPA feedstock will be sourced, typically, 50% coconut fronds (8,000 TPA) + 50% cane trash, paddy straw bales (8,000 TPA). Coconut fronds sourcing will be from KR Pet & Nagamangala Taluk in Mandya district and neighboring Chanarayapatna Taluka in Hassan district. Cane trash & Paddy straw sourcing will be from KR Pet and Pandavapura Talukas in Mandya district.

INFRASTRUCTURE

A. **Process units Building:**

Six number, each with 2 Nos 1 TPH Pyrolysis units + Infeed conveyor for pulverized biomass @ 3 TPH + 2 TPH output Dryer with up to 0.6 TPH water removal + out feed hopper with two screw conveyors + Biochar Bagging System, @ say 20 Nos 50 Kg bags per hour + individual, 1 cu m, wheeled drums for Bio Oil collection + MCC & DB's

SN	Equipment	Manufacturers specifications	Industry standards	Existing installed capacity	Forecast installed capacity	Load Factors	Efficiency

B. **Administration Building:**

This will have Administration & Accounts Staff workstations + Discussion room cum Visitors room + Security cum Reception counter + Kitchen cum Canteen + Toilets + STP

C. **Rooftop Solar + RWH system**

D. **DP Structure + 11 KV / 415 V Transformer + Electrical Room (LT Switchboards & Control Panel + O&M staff workstations) + DG Set & UPS Room**

E. **Raw Water, Treated water and Fire Reservoir & Pumping Station**

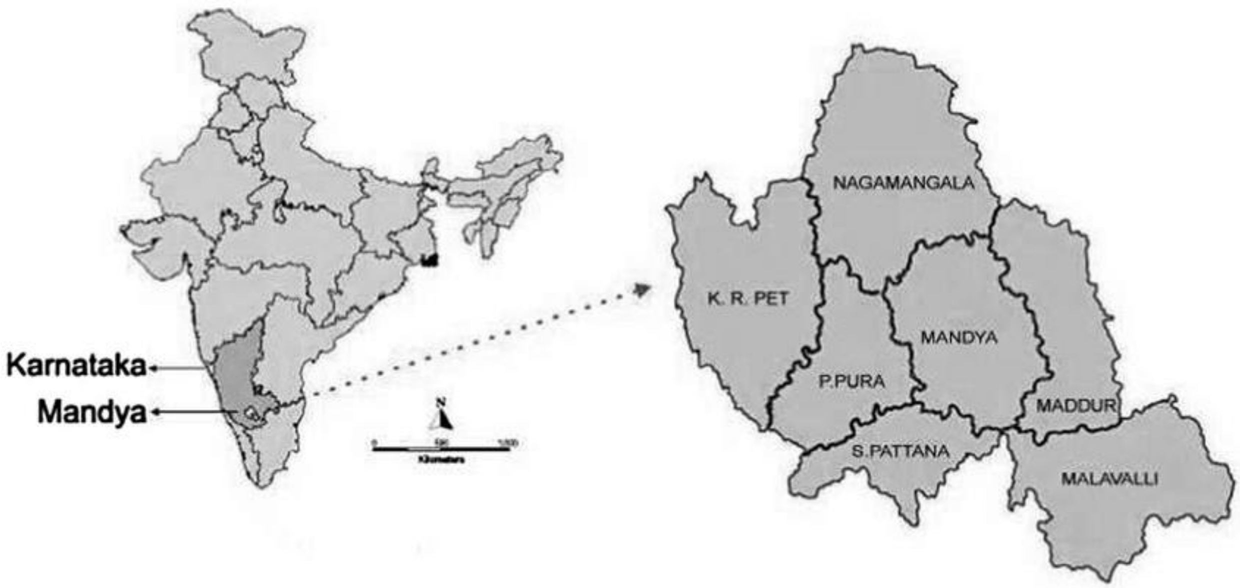
1.13 Project Location

Indicate the project location and geographic boundaries (if applicable) including a set of geodetic coordinates.

For AFOLU projects, GCS projects, grouped projects, or projects with multiple project activity instances, a separate KML file is required.

The Project is located in Bikasandra Village, Mavinakatte Koppala, Santebachenahalli of KR Pet Taluka Mandya District, Karnataka State INDIA

Lat 12.7896041 Long 76.5710126



1.14 Conditions Prior to Project Initiation

Describe the conditions existing prior to project initiation and demonstrate that the project has not been implemented to generate GHG emissions for the purpose of their subsequent reduction, removal, or destruction.

Where the baseline scenario is the same as the conditions existing prior to the project initiation, there is no need to repeat the description of the scenarios; state that this is the case and refer the reader to Section 3.4 (Baseline Scenario).

AFOLU projects must also provide the following information:

- Ecosystem type: *Provide a brief (1–2 sentence) description of the ecosystem type.*
- Current and historical land-use: *Provide a brief (2–4 sentence) description of the current and historical land use of the project area.*
- Present and prior environmental conditions of the project area: *Provide information on the climate, hydrology, topography, relevant historic conditions, soils, vegetation, and ecosystems of the project area.*

The conditions that are existing prior to the project initiation are that feedstock such as coconut fronds, cane trash, paddy straw, agricultural residue are dumped at site to natural decay and mostly burning causing harmful emissions. This indicates that project has not been implemented to generate GHG emissions and subsequent reduction, removal or destruction.

The baseline scenario is the same as the conditions existing prior to the project initiation.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Identify and demonstrate compliance of the project with all and any relevant local, regional and national laws, statutes and regulatory frameworks.

The project complies with all applicable local, state, and national laws and regulatory frameworks. Key applicable instruments and compliance status are as follows:

(1) Karnataka Factories Act, 1948 — Factory licence application to be filed with Mandya District Inspector of Factories prior to commissioning;

(2) Air (Prevention and Control of Pollution) Act, 1981 and Environment (Protection) Act, 1986 — Consent to Establish (CTE) to be obtained from KSPCB Mandya District Office prior to civil works; Consent to Operate (CTO) to be obtained prior to commissioning; stack emission limits for PM (≤50 mg/Nm^3) and CO (≤100 mg/Nm^3) to be complied with;

(3) Water (Prevention and Control of Pollution) Act, 1974 — all process and domestic effluent treated to KSPCB standards; zero liquid discharge target;

(4) EIA Notification, 2006 — pyrolysis units below 5 TPH are not scheduled activities under Schedule I; confirmation letter from SEIAA Karnataka to be obtained and attached as Appendix;

(5) Electricity Act, 2003 — HT connection (11 kV) to be obtained from BESCOM; rooftop solar registered under Karnataka Solar Policy and BESCOM net metering framework;

(6) Solid Waste Management Rules, 2016 — biochar classified as non-hazardous agricultural soil amendment; bio-oil managed as industrial by-product;

(7) Labour Laws — compliance with Minimum Wages Act 1948, Contract Labour (Regulation and Abolition) Act 1970, POSH Act 2013, and Karnataka Shops and Establishments Act;

(8) Carbon registry compliance — no parallel registration under CDM, Gold Standard, or any other GHG programme; project maintains single registry listing under Verra VCS. [Status of each approval to be updated to “Obtained” with reference number before validation is completed.]

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

If yes, provide required evidence of no double issuance as outlined by the VCS Standard.

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

If yes, provide the registration number and the date of project inactivity under the other GHG program.

Is the project active under the other program?

☐ Yes ☒ No

Project proponents, or their authorized representative, must attest that the project is no longer active in the other GHG program in the Registration Representation.

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

If yes, provide the program name(s), reason(s) and date for the rejection, justification of eligibility under the VCS Program, and any other relevant information.

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

If yes to all:

Provide evidence of the public statement. Evidence must be provided in this section or in an appendix.

1.18 Sustainable Development Contributions

Provide a brief description that includes the following (no more than 500 words):

- *A summary description of project activities that result in sustainable development (SD) contributions (i.e., technologies/measures implemented, activity location).*
- *An explanation of how project activities will result in expected SD contributions.*
- *A description of how the project contributes to achieving any nationally stated sustainable development priorities, including any provisions for monitoring and reporting these.*

The project activity contributes to sustainable development (SD) in various ways. The project being in the category of agricultural wastes management and overall process involves significant resources, the project certainly achieves a high level of SD indicators which can be monitored and reported. A summary of the overall SD contributions is given below:

Environmental Sustainability

This project leads to an effective management of waste agri-residue that helps reduce environmental pollution as in absence of the project waste agri-residue would have been degraded via open burning or open decay. Since the project implements biochar production from the waste agri-residue, thus the waste is transformed into valuable resources as soil amendments which reduces the need for chemical fertilizers, mitigates greenhouse gas emissions, and promotes soil health, biodiversity, and water conservation etc.

Resource Conservation:

This project helps utilizing the biomass wastes for a meaningful purpose rather than being discarded; it leads to significant resource conservation. By efficiently managing and utilizing these waste streams into biochar with soil application, the project reduces the demand for virgin resources, such as synthetic fertilizers. This promotes resource conservation and minimizes the pressure on natural ecosystems

Technology & knowledge sharing:

Additionally, the proposed project being first-of-its-kind in the region, it will also create awareness and knowledge transfer related to waste handling, management and production of biochar. Through initial R&D and subsequent field level experiments, testing, training etc. project will also contribute to educational engagement, qualitative and quantitative aspects for the community, beneficiaries, stakeholders etc. by skilling & developing local green entrepreneurs.

Good Health and Well Being

The project certainly improves environmental aspects, as explained above. Especially the baseline scenario of the project is attached to a high level burning if wastes in the region which is now avoided, hence directly benefits to the health of the communities due to reduction in smoke, air-particulate matters etc. Also, the waste management and systematic way of biochar production leads to a dedicated framework both in terms of resources utilization, knowledge sharing, economic upliftment, useful products and improved agricultural activities etc.; hence directly contributes to overall well-being in the region.

Rural and Economic opportunities

The proposed project envisages creating local employment and income-generation opportunities in rural areas as project activities are confined in and around the village communities. For example, setting up of biochar production units (via either of the approaches discussed under the project design) has engaged people from local communities and contributed to the local economy by means of adding earning to the households. Such economic-benefits attached to the project are gender un-biased, which means both male and female are getting direct and indirect employment across various functions of the project, leading to gender equality as well. Also, the end product from the project is Biochar which will be targeted for soil-applications; thus, the proposed project helps turning agricultural wastes into marketable products like organic fertilizers, farmers can generate additional income and improve their livelihoods.

Soil Health and Fertility

Biochar has beneficial effects on soil health and fertility. It improves soil structure, water holding capacity, and nutrient retention. By enhancing soil health, biochar can increase agricultural productivity, reduce the need for synthetic / chemical fertilizers, and decrease nutrient runoff, which helps prevent water pollution and eutrophication. Since biochar-amended soils have the potential to improve water quality by reducing nutrient leaching and runoff, it can directly impact life above the ground as well as below water as it can help retain and slowly release nutrients, minimizing their movement into water bodies. This helps protect aquatic ecosystems, prevent water pollution, and maintain water quality for both human consumption and biodiversity conservation.

Climate Change Mitigation Adaptation

The climate change mitigation is one of the key integral benefits attached to this project. Biochar is a stable form of carbon that can be incorporated into soil. By adding biochar to agricultural lands, carbon is sequestered for long- lived periods, effectively removing carbon dioxide from the atmosphere. This process helps mitigate climate change by reducing greenhouse gas emissions and offsetting carbon footprints. Additionally, the sustainable management of agricultural waste can contribute to climate change mitigation by reducing greenhouse gas emissions associated with the baseline practices such as open burning, open decay etc. Also as already mentioned in the previous para, this sustainable waste management practices improve soil fertility and water management, enhancing resilience to climate change impacts for long term.

Circular Economy approach

This project can be attributed to embrace a circular economy approach by closing the loop in the waste management system. Conventionally, the waste agri-residue is being considered as a burden, being brunt without the purpose driven approach. Whereas under the project level the waste is now seen/realized as a valuable resource that can be recycled and reused, i.e. production of biochar which goes for soil applications to enable various positive gains. Thus, by integrating waste input into the overall waste management system, the project maximizes resource efficiency to result in a useful product and minimizes waste generation in the region.

Overall, it can be summarized that the project contributes to sustainable development by promoting various aspects including (but not limited to) environmental conservation, resource efficiency, rural development, climate change mitigation, and the transition towards a circular economy. It aligns agricultural practices with the principles of sustainable development, fostering a more resilient and equitable agricultural sector.

Therefore, PP attributes the following SDG indicators in this project activity:

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Where applicable, describe the leakage management plan and implementation of leakage and risk mitigation measures.

The leakage management is ensured by providing Captive Rooftop Solar power plant of approx. 100 KW with BESS installed at the proposed biochar plant.

Internal mobility and transport Utility will comprise of EV transport vehicles.

1.19.2 Commercially Sensitive Information

Indicate whether any commercially sensitive information has been excluded from the public version of the project description using Appendix 1, and briefly describe the items to which such information pertains. Provide justification for why the information is commercially sensitive and confirm that it is not otherwise publicly available.

Note - Information related to the determination of the baseline scenario, demonstration of additionality, and estimation and monitoring of GHG emission reductions and removals (including operational and capital expenditures) cannot be considered to be commercially sensitive and must be provided in the public versions of the project documents.

Not Applicable – no commercially sensitive information has been excluded

1.19.3 Further Information

Include any additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the GHG emission reductions or carbon dioxide removals, or the quantification of the project's reductions or removals.

Not Applicable –the selected project site is not impacted by the legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Use the table below to describe the stakeholder identification process. Where the rows do not apply, provide justification in the cell in the table below.

Stakeholder Identification	*Describe the process(es) used to identify stakeholders likely impacted by the project. List the stakeholders identified.* • Public meetings will be arranged and the project benefits will be highlighted to the participants, stake holders showing interest will then be identified for the engagement and consultation • Farmers and Village officials

Legal or customary tenure/access rights	*Describe any legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, Indigenous People (IPs), local communities (LCs), and customary rights holders.* • PP owns the land intended for the biochar plant
Stakeholder diversity and changes over time	*Describe the social, economic, and cultural diversity within stakeholder groups, the differences and interactions between the stakeholder groups, and any changes in the make-up of each group over time.* • The stakeholders are farmers with varying sizes of land holdings with agriculture income • The social, economic, and cultural diversity may vary between the stakeholder groups as it depends on the size of the family, immediate relatives living in the village, the land parcel they own, the output of the produce in their field etc.
Expected changes in well-being	*Describe the expected changes in well-being and other stakeholder characteristics relative to the baseline scenario, including changes to ecosystem services identified as important to stakeholders;* • User of biochar for soil amendment ensures better crop output resulting in crop revenue increase. • Skill development on green job creates entrepreneurship opportunity, improved social wellbeing, poverty alleviation and support climate initiatives.
Location of stakeholders	*Describe the location of stakeholders, IPs, LCs, and customary rights holders, and areas outside the project area that are predicted to be impacted by the project.* • Stakeholders are located around the proposed project location which is owned by the PP
Location of resources	*Describe the location of territories and resources which stakeholders own or to which they have customary access.* • Stakeholders reside in K R Pet, Pandavapura, Nagamangala & Channarayapatna. • They own land, tractors, tillers, farm equipment & implements

2.1.2 Stakeholder Consultation and Ongoing Communication

Use the table below to describe the process for and the outcomes from the stakeholder consultation conducted prior to project initiation.

Date of stakeholder consultation	*DD-Month-YYYY* 01-March-2026 Planned
Stakeholder engagement process	*Describe the process to engage stakeholders in a culturally appropriate manner (e.g., dates of announcements or meetings, language and gender sensitivity). Describe the process or methods used to document the outcomes.* • The public meeting will be announced in advanced through the cooperative & self-help group connects. • The meetings, discussions & interactions will the through audio-visual aids & will be in the local language. • The topics will be gender neutral. • Interviews will be conducted and the interaction outcome will be documented in a questionnaire.
Consultation outcome	*Summarize the discussion around consent to project design and implementation, risks, costs and benefits of the project, all relevant laws and regulations covering workers' rights in the host country, the discussion of FPIC and the VCS validation and verification process.* • Every respondents' response will be recorded in a questionnaire and consent obtained for the project design & implementation.
Ongoing communication	*Describe the mechanisms for ongoing communication with stakeholders.* • Regular meetings & periodic consultation sessions will be arranged to keep the communication ongoing.
Stakeholder input	*Describe how due account was taken of all input received during the consultation. Include details on any updates to the project design or justify why updates were not necessary or appropriate.* • The project updates will be shared during the regular meetings & periodic consultation sessions.

2.1.3 Free Prior and Informed Consent

Use the table below to describe the outcome of the FPIC process as part of the stakeholder consultation process.

Obtaining consent	*Describe and demonstrate how consent to implement the project activities was obtained from those concerned, including IPs, LCs, and customary rights holders, and a transparent agreement was reached. Describe any ongoing or unresolved conflicts and demonstrate that the project does not exacerbate nor influence the outcomes of unresolved conflicts.* • PP owns the land where the biochar plant will be setup, hence consent to implement the project is not required.
Outcome of FPIC	*Describe the outcome of the FPIC process, the transparent agreement, and the information disclosed prior to establishing a transparent agreement with those concerned, IPs, LCs, and customary rights holders. Provide assurance that the project has not encroached on land, relocated people without consent, and forced physical or economic displacement.* • PP owns the land hence encroachment on land & relocation of people arise.

2.1.4 Grievance Redress Procedure

Use the table below to describe the grievance redress procedures developed to resolve any conflicts which may arise between the project proponent and stakeholders.

Development process	*Describe the process used to develop the grievance redress procedure including processes for receiving, hearing, responding and attempting to resolve grievances within a reasonable time period, taking into account culturally appropriate conflict resolution methods.* • The village panchayat office help will be sought to ensure all grievance are collected, recorded & shared with the PP for receiving, hearing, responding & addressal within a stipulated time frame.
Grievance redress procedure	*Describe the grievance redress procedures developed with stakeholders.* • Impacted stakeholders will be reached immediately over phone to understand their grievance & to address them

	• Where required a physical meeting will be arranged with the impacted stakeholders to understand their grievance & to address them

2.1.5 Public Comments

Summarize any public comments submitted during the public comment period and any comments received after the public comment period. Demonstrate how due account was taken of all comments received. Include details on when the comments were received, and any updates to the project design or demonstrate the insignificance or irrelevance of comments.

Comments received	Actions taken
Summary of comment received	*Provide a summary of actions taken and any project design updates or justify why updates were not necessary or appropriate.*
Yet to start	Yet to start

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

Demonstrate that management teams have expertise or experience in implementing similar project activities and engaging communities. Where relevant experience is lacking, demonstrate how the project proponent has partnered with other organizations to support the project or have a recruitment strategy to fill the identified gaps.

The Management team is a highly qualified and a richly experienced team with each associate of the organisation having experience of more than 3 decades in technology and management related to environment management and sustainability. The PP has the experience of successfully setting up & operating 4.5 MW Malavalli Power Plant, world's 1st commercial scale bioenergy plant based on low density crop residues, which also delivered the world's 1st Gold Standard CER's.

2.2.2 Risk Assessment

Use the table below to describe the risk assessment and outcome of the potential risks to stakeholders and the environment. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	Air & Water pollution.	Educating the stakeholders about the health hazards of waste agri-residue burning in the fields and leaving them to decay. Showcase the benefits of biochar and the health benefits it offers.
Risks to stakeholder participation	Non participation.	Ensure participation with incentives.
Working conditions	Not favorable & unsafe working condition.	Provide a favorable & safe work environment
Safety of women and girls	Discrimination, harassment & unsafe working condition.	Conduct sensitivity sessions to educate men folks about the safety & respect towards women & girls. Ensure a safe working environment.
Safety of minority and marginalized groups, including children	Discrimination & harassment.	Conduct sensitivity sessions to educate the participants about the safety & respect towards others.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Release of pollutants from the biochar plant.	Pollution control systems like air filters & water treatment plant will be installed to monitor & prevent any leakage.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Use the table below to identify and summarize the risks for rights related to labor and work. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Discrimination	Physical & Emotional trauma.	Conduct sensitivity sessions to educate the participants about the empathy & respect towards others. Equal opportunity will be practiced.
Sexual harassment	Physical & Emotional trauma.	Conduct sensitivity sessions to educate men folks about the safety & respect towards women & girls.
Equal pay for equal work	Discrimination.	Skill development & skilling is done by the PP to ensure technicians get equal pay for equal work.
Gender equity in labor and work	Discrimination.	Gender neutrality & equity will be followed.
Forced labor	Discrimination.	All labors will be employed with consent. Strict adherence to the local government's statutory compliance.
Child labor	No risk identified.	Child labor is strictly prohibited.
Human trafficking	No risk identified.	Only locals trained & skilled will be employed in the project.

2.3.2 Human Rights

Use the table below to identify and summarize any risks related to recognizing, respecting, and promoting the protection of the rights of IPs, LCs, and customary rights holders in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	Only locals and stakeholders identified, trained & skilled will be employed in the project. This

	will ensure that the human rights are not violated & equality is maintained.

2.3.3 Indigenous Peoples and Cultural Heritage

Use the table below to identify and summarize any risks related to recognizing, respecting, and promoting the protection of the rights of IPs, LCs, and customary rights holders, and to preserving and protecting cultural heritage as part of project activities. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

Risks identified	Mitigation(s) or preventative measure taken
No risk identified.	Only locals and stakeholders identified, trained & skilled will be employed in the project. This will ensure that the local cultural heritage is preserved & any external influence is prevented.

2.3.4 Property Rights

Use the table below to identify and summarize any risks related to protecting and preserving the property rights of IPs, LCs, and customary rights holders, and to protecting legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

Risks identified	Mitigation or preventative measure(s) taken
No risk identified.	• PP owns the land where the proposed biochar plant will be setup. • Wherever land is required to be taken on lease, a well-documented lease agreement will be signed mutually between the PP & the land owners. This will have no impact on the property rights of the land owner.

2.3.5 Benefit Sharing

Where the project impacts property rights as described in Section 2.4.4 above, use the table below to describe the project's benefit sharing agreement.

Process used to design the benefit sharing plan	*Describe the process used to develop the benefit-sharing agreement with the affected stakeholder groups.* Barter system of Biochar to waste agri-residue collected from the fields.
Summary of the benefit sharing plan	*Describe the benefit-sharing agreement. Where affected stakeholder groups wish to keep elements of the plan private, provide the full arrangement as a commercially sensitive document. The project proponent shall demonstrate that the community wishes to keep this information private.* Exchange of Biochar against the waste agri-residue collected from the fields.
Approval and dissemination of benefit sharing plan	*Demonstrate that the benefit- sharing agreement was agreed up on by the affected stakeholder groups, and that the agreement was shared in a culturally appropriate manner. Demonstrate that the agreement is readily accessible should stakeholders wish to review the agreement.* The agreement will be shared with the stakeholders for ready reference.

2.4 Ecosystem Health

Use the table below to identify and summarize any risks related to impacts on biodiversity and ecosystems, soil degradation and soil erosion, and water consumption and stress. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified.	All preventive measures to protect the biodiversity & ecosystem are considered prior to the setting up of the biochar plant.

Soil degradation and soil erosion	No risk identified.	All preventive measures are considered prior to the setting up of the biochar plant.
Water consumption and stress	Water shortage.	Rainwater harvesting, raw water treatment & storage is considered as part of the biochar plant.

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☒ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	*Demonstrate that the project will not adversely impact habitats and areas needed for habitat connectivity for rare, threatened, or endangered species.*
Areas needed for habitat connectivity	*Demonstrate that the project will not adversely impact areas needed for habitat connectivity.*
…	…

Use the table below to identify and summarize any risks related to habitats for rare, threatened, and endangered species, and for areas for habitat connectivity. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	No risk identified.	Biochar plant is not located adjacent to habitats for rare, threatened, or endangered species
Areas for habitat connectivity	No risk identified.	Biochar plant location does not hinder any habitat connectivity

2.4.2 Introduction of Species

Demonstrate, using the table below, that no invasive species will be used as part of project activities. Categorize each species as native, non-native, and indicate if the species is a monoculture. Where the species is non-native, include an explanation of possible adverse *effects of its use and a description of how the project will mitigate such risks. For projects with no planting or species introduction, this section may be indicated as N/A.*

Species introduced	Classification	Justification for use	Adverse effects and mitigation
N/A			

Where invasive species exist in the project area, list such species in the table below and describe the commensurate mitigation measure(s) in place to prevent the spread or continued existence of invasive species.

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
N/A	

Use the table below to identify and summarize any risks related to invasive species. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species	No risk identified.	If any non-native invasive species is identified immediate corrective action will be initiated with the help of the forest department

2.4.3 Ecosystem Conversion

ARR, ALM, WRC or ACoGS projects shall provide evidence that the project area was not cleared or drained of existing natural ecosystems, unless such clearing took place at least 10 years prior, or the dominant land cover was invasive.

Use the table below to identify and summarize any risks related to ecosystem conversion. Describe the commensurate mitigation or preventative measure(s) in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column. Add rows as needed.

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified.	The identified project area is agricultural land & no existing natural ecosystem was cleared or drained in the past 10 years.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Provide the title, reference and version number of the following information for the methodology(s), tools, and modules applied to the project (where applicable).

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0044	VCS Methodology	1.2

3.2 Applicability of Methodology

Demonstrate and justify how the project activity(s) meets each of the applicability conditions of the methodology(s), tools, and modules applied by the project (where applicable). Address each applicability condition separately.

Methodology ID	Applicability condition	Justification of compliance
VM0044	*First applicability condition for given methodology, tool, or module*	*Justification that the project complies with this applicability condition*

3.3 Project Boundary

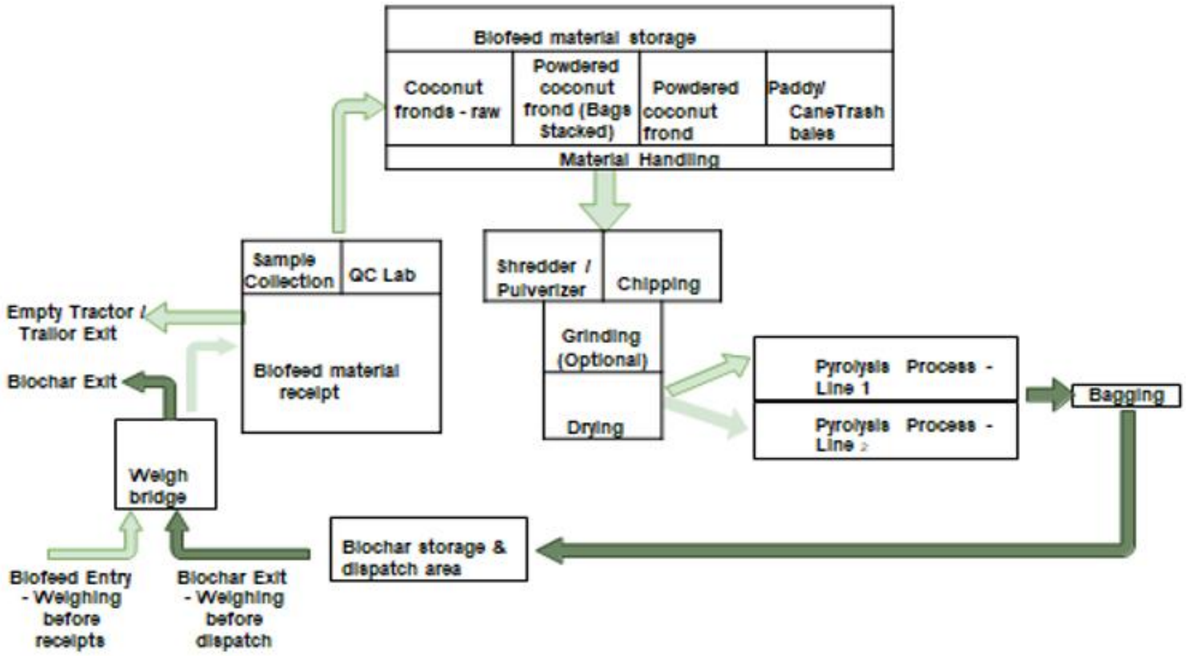
Define the project boundary and identify the relevant GHG sources, sinks and reservoirs for the project and baseline scenarios (including leakage if applicable). Add rows as needed.

Source		Gas	Included?	Justification/Explanation
Baseline	Feedstock production	CO_2	No	Excluded. Waste biomass is considered renewable as per eligibility
Baseline	Feedstock production	CH_4	No	
Baseline	Feedstock production	N_2O	No	
Baseline	Feedstock production	Other	No	
Baseline	Feedstock Transportation	CO_2	No	Expected to be de minimis if distance between sourcing sites and production facility is less than 200 kilometers.
Baseline	Feedstock Transportation	CH_4	No	
Baseline	Feedstock Transportation	N_2O	No	
Baseline	Feedstock Transportation	Other	No	
Project	Combustion, aerobic and anaerobic decomposition of feedstocks	CO_2	No	Possible emissions from decay or combustion of biomass in the absence of project activity are excluded. Baseline emissions are assumed to be zero (a conservative assumption).
Project	Combustion, aerobic and anaerobic decomposition of feedstocks	CH_4	No	
Project	Combustion, aerobic and anaerobic decomposition of feedstocks	N_2O	No	
Project	Combustion, aerobic and anaerobic decomposition of feedstocks	Other	No	
Project	Pyrolysis (High technology systems)	CO_2	No	High technology systems are provided a default emission value
Project	Pyrolysis (High technology systems)	CH_4	Yes	
Project	Pyrolysis (High technology systems)	N_2O		
Project	Pyrolysis (High technology systems)	Other		
Project	Electricity and/or fossil fuels consumed during eligible thermochemical process	CO_2	Yes	Included. Emissions associated directly with project activity due to use of fossil fuel.
Project	Electricity and/or fossil fuels consumed during eligible thermochemical process	CH_4	No	
Project	Electricity and/or fossil fuels consumed during eligible thermochemical process	N_2O	No	
Project	Electricity and/or fossil fuels consumed during eligible thermochemical process	Other	No	
Project	Biochar transportation	CO2	No	Expected to be minimal if distance between production facility and end-use destination is less than 200 kilometers. Usage of EV

Source		Gas	Included?	Justification/Explanation
		CH4	No	
		N20	No	
		Other	No	
	Pre-treatment of feed stocks (e.g., grinding, drying)	CO2	No	
		CH4	No	
		N20	No	
		Other	No	

Provide a diagram or map of the project boundary, showing clearly the physical locations of the various installations or management activities taking place as part of the project activity based on the description provided in Section 1.12 (Description of the Project Activity) above.

For non-AFOLU projects, include in the diagram the equipment, systems and flows of mass and energy. Include the GHG emission sources identified in the project boundary.

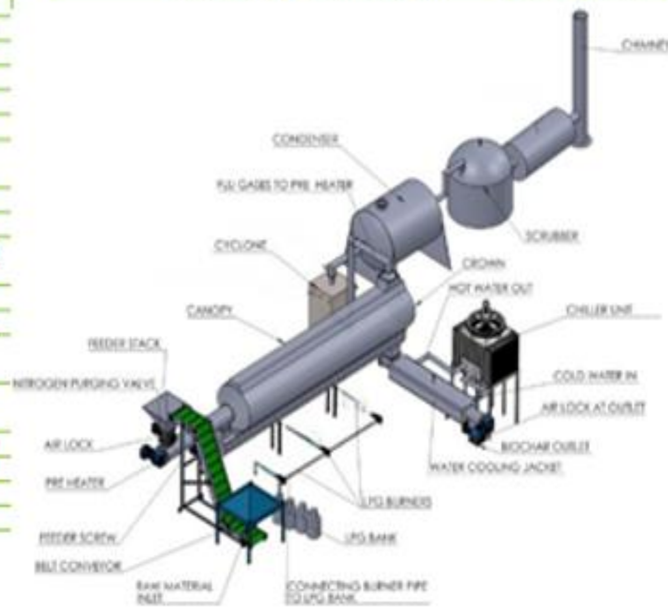


Biofeed material storage
Coconut fronds - raw
Powdered coconut frond (Bags Stacked)
Powdered coconut frond
Paddy/ CaneTrash bales
Material Handling
Sample Collection
QC Lab
Biofeed material receipt
Empty Tractor / Trailor Exit
Biochar Exit
Weigh bridge
Biofeed Entry - Weighing before receipts
Biochar Exit - Weighing before dispatch
Shredder / Pulverizer
Chipping
Grinding (Optional)
Drying
Pyrolysis Process - Line 1
Pyrolysis Process - Line 2
Bagging
Biochar storage & dispatch area

Equipment: Continuous Biochar Machine

Technical Specifications: Continuous Type			
PART	Particulars / Dia mentions / Specifications	Power	Qty
Feeder Conveyor	12 feet Feeder Bucket Conveyor with base and top hopper	3 HP	1 No
	3 HP, 3 phase motor, 1880 rpm		
Input Airlock	Airlock above main reactor inlet, 1 HP motor	1 HP	1 No
Pre-Heater	4 mt long Screw Conveyor	3 HP	1 No
	Flu gas heating from main reactor		
	3 HP, 3 phase geared motor		
	3 Temp sensors		
	Outer jacket with flu gas from main reactor		
	LRB Insulation with GI Cladding		
	ID Fan for flu gases		
Main Reactor	5 mt long Reactor	5 HP	1 No
	Twin Screw Conveyor		
	LPG Gas Heating and Syngas heating burners		
	5 HP, 3 phase geared motor		
	5 Temp sensors		
	Outer Chamber with LRB Insulation and GI Cladding		
	8 to 10 Syngas burners below main reactor		
	6 TO 8 LPG Burners for cold start operation		
Condensers	2 nos 150 mm diameter x 2 mt length Condensers with water cooling		2 Nos
	Bio Oil Collection Tanks 2 nos, 0.5 x 0.5 x 0.5 m		
Charcoal Outlet	4 mt long Screw Conveyor	3 HP	1 No
	Outer jacket with cooled water flow from Chiller		
	3 temp sensors for inlet and outlet temp sensing		
Airlock Outlet	Airlock for outlet of Biochar with 1 HP motor	1.0 HP	1 No
Chiller Unit	50 TR Chiller Unit with water tank	10 HP	1 no
	Associated GI piping		
	3 Phase 440 volts supply		

Technical Specifications: Continuous Type			
Control Panel	VFD Speed control of all 3 conveyors	1 HP	1 No
	PRE-Heater, MAIN Reactor, COOLER Unit		
	Digital Temp Display for PRE, MAIN and COOLER screws		
	Chiller Controller		
Pipes	Pipes for Heated gas and Syn-gases		lot
Miscellaneous	Minor parts, machines, spares, nut bolts, watch windows etc for running above plant		lot
Scrubber	Syngas scrubber for filtration and extraction after Condensers		1 No
	Outlet for particulate matter and oil at bottom		
Chimney	150 mm diameter, 6 ft height MS Chimney		1 No
	For heated gases above pre-heater		
	TOTAL ELECTRICITY CONSUMPTION	26 HP	20 kW



For AFOLU projects, include in the diagram or map the locations of where the various measures are taking place, any reference areas and leakage belts.

N/A

3.4 Baseline Scenario

Identify and justify the baseline scenario, in accordance with the procedure set out in the applied methodology and any relevant tools. Where the procedure in the applied methodology involves several steps, describe how each step is applied and clearly document the outcome of each step.

Explain and justify key assumptions, rationale, and methodological choices. Provide all relevant references.

The baseline scenario is that in which, in the absence of the project activity, the coconut fronds, cane trash, paddy straw and waste agri-residue biomass is mainly combusted without any utilization for purposes and is not utilized for producing biochar for either soil or non-soil application.

As per the applicability conditions of the methodology, the waste biomass must only have the following fates: decay (aerobic or anaerobic) or combustion of biomass for purposes other than energy production. The resulting emission avoidance potential for the project activity feedstock has been excluded (a conservative assumption).

In this regard, a baseline survey was conducted and it has been confirmed that the baseline scenario has the fate as combustion without energy production hence excluded hence baseline scenario of the project is in line with the requirement of the applied methodology. Also, emissions avoidance associated with the feedstock at the baseline is excluded.

3.5 Additionality

Demonstrate and assess the additionality of the project, in accordance with the applied methodology and any relevant tools, taking into account the following additionality methods:

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☒ No

If no, describe which mandated laws, statutes, or other regulatory frameworks require project activities and provide evidence of systematic non-enforcement to demonstrate regulatory surplus.

3.5.2 Additionality Methods

- *Where a project method is applied to demonstrate additionality and the procedure in the applied methodology or tool involves several steps, describe how each step is applied and clearly document the outcome of each step. Indicate clearly the method selected to demonstrate additionality (e.g., investment analysis or barrier analysis in the case of the CDM Tool for the demonstration and assessment of additionality). Where barrier analysis, or equivalent, is used to demonstrate additionality, only include the most relevant barriers. Justify the credibility of the barriers with key facts and/or assumptions and the rationale. Provide all relevant references.*

- *Where a performance method is applied to demonstrate additionality, demonstrate that performance can be achieved to a level at least equivalent to the performance benchmark metric.*
- *Where the methodology applies an activity method for the demonstration of additionality, include a statement that notes that conformance with the positive list is demonstrated in the Applicability of Methodology section above.*

- *Provide sufficient information (including all relevant data and parameters, with sources) so that a reader can reproduce the additionality analysis and obtain the same results.*

- Regulatory Surplus

The project is located in India, a Non-Annex 1 country under the UNFCCC. There is no national, state, or local law, statute, or regulatory framework that mandates the conversion of agricultural residues into biochar or requires carbon dioxide removal activities of this nature. While Karnataka has issued directives discouraging open field burning of crop residues (Karnataka Prevention and Control of Pollution Act; KSPCB Circular on Stubble Burning, 2022), these directives are not systematically enforced, as evidenced by persistent open burning documented by NASA FIRMS satellite fire detection data for Mandya and Hassan districts during the 2022–2024 crop residue seasons (see Appendix 2 — Baseline Survey Report). The project therefore demonstrates regulatory surplus in accordance with the VCS Standard.

- Additionality Demonstration — Activity Method (Positive List)

VM0044 v1.2 Section 6.2 states that biochar production from waste biomass qualifies as an activity included on Verra's Positive List of additional project activities. The following table confirms that all conditions for applying the Activity Method are met by this project:

Condition 1 — Non-Annex 1 country: CONFIRMED. India is a Non-Annex 1 Party to the UNFCCC.

Condition 2 — Feedstock is waste biomass (not purpose-grown): CONFIRMED. Coconut fronds, cane trash, and paddy straw are agricultural residues generated as by-products of primary crop cultivation in Mandya and Hassan districts. None of these materials are grown specifically for energy or biochar production.

Condition 3 — Project activity is not legally mandated: CONFIRMED. No applicable law requires biochar CDR production in India, Karnataka, or Mandya District.

Condition 4 — Project activity is not common practice in the project region: CONFIRMED. There is no existing commercial-scale biochar CDR facility operational in Karnataka or the wider South India region as of the project start date. The Malavalli Power Plant (biomass gasification, 4.5 MW) is the only comparable biomass-to-energy precedent in Mandya district; biochar CDR via continuous pyrolysis is a materially different activity not replicated in the region. The project therefore demonstrates additionality through the Activity Method. No investment analysis or barrier analysis is required under VM0044 v1.2 when the Positive List criterion is satisfied. The conformance with the Positive List is further demonstrated in Section 3.2 (Applicability of Methodology) above, in which each VM0044 applicability condition is addressed and confirmed.

3.6 Methodology Deviations

Describe and justify any methodology deviations applied, including any previous deviations. Include evidence to demonstrate the following:

- *The deviation will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals.*
- *The deviation relates only to the criteria and procedures for monitoring or measurement and does not relate to any other part of the methodology.*

No deviations applied.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Describe the procedure for quantification of baseline emissions and/or carbon stock changes in accordance with the applied methodology. Baseline emissions may be negative where carbon stock increases (sinks) exceed baseline emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

$BE_{SS,y}$ = “0” following the conservative approach of VM0044 for waste biomass disposal.

4.2 Project Emissions

Describe the procedure for quantification of project emissions and/or carbon stock changes in accordance with the applied methodology. Project emissions may be negative where carbon *stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.*

This is the Project Emissions "PE_y" (Footprint) of biochar modules.

$$PE_{PS,py} = (P_{ED,p,y} + P_{EP,p,y} + P_{EC,p,y}) \times (\sum_t \sum_k M_{t,k,p,y}/M_{p,y})$$

$P_{ED,p,y}$ = Emission associated with pre-treatment of biomass (Hammer Mill @133HP * 24 hrs * 6 days * 48 weeks)

$P_{EP,p,y}$ = Emission associated with conversion of biomass (high technology it is "0")

$P_{EC,p,y}$ = Emissions due to the utilization of auxiliary energy (Pyroliser - 52HP * 24 hrs + Bagging - 5HP * 24hrs + Lighting - 7HP * 24hrs)

$M_{t,k,p,y}$ = Mass on a dry weight basis of biochar

$M_{p,y}$ = Total mass of biochar on a dry weight basis

$PE_{PS,py}$	=	725.09	tCO_2e
$P_{ED,p,y}$	=	489.53	tCO_2e
$P_{EP,p,y}$	=	0	tCO_2e
$P_{EC,p,y}$	=	235.56	tCO_2e
$M_{t,k,p,y}$	=	5200	tons
$M_{p,y}$	=	5200	tons

4.3 Leakage Emissions

Describe the procedure for quantification of leakage emissions in accordance with the applied methodology. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

Leakage emission is considered “0”

(VCS Methodology VM0044 - Section 8.3)

*These are emissions **outside** your facility boundary.*

- ***Transport Emissions:***
  - *VM0044 provides a **threshold:** Transport emissions for feedstock and biochar are **only** counted if the transport distance exceeds **200 km**.*
  - *If your collection radius is <200 km, this deduction is **zero** under VM0044.*

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Describe the procedure for the quantification of estimated GHG emission reductions (reductions) and carbon dioxide removals (removals). Include all relevant equations.

For data and parameters monitored, use the estimated data/parameter values provided in Section 5.2 below. Document how each equation is applied in a manner that enables the reader to reproduce the calculations. Provide calculations for all key equations to allow the reader to reproduce the annual calculations for estimated reductions or removals. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all of the above in the emission reduction and removal calculation spreadsheet.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCUs for reductions and removals separately where the applied methodology provides procedures and equations to do so.

For projects that are not required to assess permanence risk, complete the table below for the project crediting period:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)
DD-MMM-YYYY to 31-Dec-YYYY	*Example: 50,000*	*Example: 20,000*	*Example: 10,000*	*Example: 10,000*	*Example: 10,000*	*Example: 20,000*
01-Sep-2026 to 31-Mar-2027	*0*	*725.09*	*0*	-423	*4,799*	*4,376*

01-Apr-2027 to 31-Mar-2028	*0*	*2175.26*	*0*	-1088	*12,341*	*11,253*
01-Apr-2028 to 31-Mar-2029	*0*	*3625.43*	*0*	-1813	*20,568*	*18,755*
01-Apr-2029 to 31-Mar-2030	*0*	*4350.52*	*0*	-2175	*26,895*	*24,720*
01-Apr-2030 to 31-Mar-2031	*0*	*4350.52*	*0*	-2175	*26,895*	*24,720*
01-Apr-2031 to 31-Mar-2032	*0*	*4350.52*	*0*	-2175	*26,895*	*24,720*
01-Apr-2032 to 31-Mar-2033	*0*	*4350.52*	*0*	-2175	*26,895*	*24,720*
01-Apr-2033 to 31-Mar-2034	*0*	*4350.52*	*0*	-2175	*26,895*	*24,720*
01-Apr-2034 to 31-Mar-2035	*0*	*4350.52*	*0*	-2175	*26,895*	*24,720*
01-Apr-2035 to 31-Mar-2036	*0*	*4350.52*	*0*	-2175	*26,895*	*24,720*
Total	0	18,549	0	-18,549	0	0

For projects required to assess permanence risk:

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	*Example: 20%*
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO2e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.

State, in tCO_2e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☐ No If no, provide justification.

ii) Complete the table below for the project crediting period. Note that the buffer pool allocation is split proportionally between the estimated reductions and removals. (For example, if a project is estimated to achieve 20,000 tCO_2e removals and 80,000 tCO_2e reductions and has a buffer contribution of 20%, or 20,000, the estimated removal VCUs would be 16,000 and reduction VCUs would be 64,000).

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated buffer pool allocation (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCU issuance (tCO_2e)
DD-MMM-YYYY to 31-Dec-YYYY	*Example: 50,000*	*Example: 20,000*	*Example: 10,000*	*Example: 4,000*	*Example: 8,000*	*Example: 8,000*	*Example: 16,000*
01-Jan-YYYY to 31-Dec-YYYY							
01-Jan-YYYY to DD-MMM-YYYY							
Total							

5 MONITORING

5.1 Data and Parameters Available at Validation

Complete the table below for all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period (copy the table as necessary for each data/parameter). The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above. Data and *parameters to be monitored during the operation of the project are included in Section 5.2 (Data and Parameters Monitored) below.*

A representative sample of biochar shall be collected from each modular unit for every 10 tonnes of production. Samples will be analyzed by an ISO 17025 accredited laboratory using ASTM D5373 (for C, H, N) and ASTM D1762-84 (for moisture/ash) to confirm the molar $H:C_{org}$ ratio remains ≤ 0.7"

Data / Parameter	Biochar Quality & Stability
Data unit	*Degrees Celsius (°C)*
Description	*Provide a brief description of the data/parameter* Molar $H:C_{org}$ ratio and Organic Carbon content ($f_{C,y}$) Sampling must be done per "batch" or at a defined interval (e.g., every 500 tonnes produced per module).
Source of data	*Type K thermocouples installed in pyrolysis chamber and afterburner; data logged by PLC at 15-minute intervals to on-site SCADA and mirrored to digital MRV cloud platform.*
Value applied	*fC,y = 0.60 (conservative reference); H:Corg ≤ 0.40 (target); PV100 = 0.89 (VM0044 Table 1). To be replaced with measured values at first verification using IBI-certified laboratory results.*
Justification of choice of data or description of measurement methods and procedures applied	*Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.*
Purpose of data	*Indicate one of the following:* • *Determination of baseline scenario (AFOLU projects only)* • *Calculation of baseline emissions* • *Calculation of project emissions* • *Calculation of leakage*
Comments	*Provide any additional comments*

5.2 Data and Parameters Monitored

Complete the table below for all data and parameters that will be monitored during the project crediting period (copy the table as necessary for each data/parameter). The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 4 above.

The weight of biochar produced by each module will be recorded using calibrated digital scales. Moisture content will be measured for each batch to convert 'as-received' weight into 'dry mass' $M_{BC,y}$ as required by the VM0044 equations.

Data / Parameter	Production Quantity ($M_{BC,y}$)
Data unit	*Indicate the unit of measure*
Description	*Provide a brief description of the data/parameter* Dry mass of biochar
Source of data	*Indicate the source(s) of data*
Description of measurement methods and procedures to be applied	*Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).*
Frequency of monitoring/recording	*Continuous; logged every 15 minutes. Daily summary (min/max/mean per module) generated by SCADA and transmitted to MRV platform.*
Value applied	*Provide an estimated value for the data/parameter*
Monitoring equipment	*Type K thermocouples, IEC 60584-1 Class 1 (accuracy ±1.5 °C or 0.4%, whichever is greater); Sanjivani Agro-supplied PLC with data logger. Serial numbers to be recorded at commissioning.*
QA/QC procedures to be applied	*Thermocouples calibrated annually against NABL-traceable reference. Any production period during which afterburner temperature falls below 850 °C is automatically flagged; default methane emission factor applied per VM0044 for that batch. SCADA alert triggers manual inspection if temperature deviation persists for more than 30 minutes.*
Purpose of data	*Indicate one of the following:* • *Calculation of baseline emissions* • *Calculation of project emissions* • *Calculation of leakage*

Calculation method	*Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.*
Comments	*Provide any additional comments*

Each modular unit is equipped with thermocouples to continuously monitor the pyrolysis chamber and afterburner temperatures. Automated data loggers will record temperatures at 15-minute intervals. Any production occurring while the afterburner is below 850 °C will be subject to the default methane emission factors prescribed in VM0044

Data / Parameter	Pyrolysis temperature and Afterburner/Flare temperature
Data unit	*Indicate the unit of measure*
Description	*Provide a brief description of the data/parameter* To justify low methane (CH_4) emissions
Source of data	*Indicate the source(s) of data*
Description of measurement methods and procedures to be applied	*Specify the measurement methods and procedures, any standards or protocols to be followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).*
Frequency of monitoring/recording	*Specify measurement and recording frequency*
Value applied	*Provide an estimated value for the data/parameter*
Monitoring equipment	*Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.*
QA/QC procedures to be applied	*Describe the quality assurance and quality control (QA/QC) procedures to be applied, including the calibration procedures where applicable.*
Purpose of data	*Indicate one of the following:* • *Calculation of baseline emissions* • *Calculation of project emissions* • *Calculation of leakage*
Calculation method	*Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.*

Comments	*Provide any additional comments*

Chain of Custody (End-Use Verification)

To prove the biochar was actually applied to soil (or an eligible non-soil use).

Data to Collect	Verification Method
Delivery Point	GPS coordinates of the farm/application site.
Application Rate	Delivery notes signed by the end-user.
Non-Soil Use	Certificates of incorporation (e.g., in concrete or asphalt).

5.3 Monitoring Plan

Describe the process and schedule for obtaining, recording, compiling and analyzing the monitored data and parameters set out in Section 5.2 (Data and Parameters Monitored) above.

Include details on the following:

- *The methods for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters. Where relevant, include the procedures for calibrating monitoring equipment.*
- *The organizational structure, responsibilities and competencies of the personnel that will be carrying out monitoring activities.*
- *The procedures for internal auditing and QA/QC.*
- *The procedures for handling non-conformances with the validated monitoring plan.*
- *Any sampling approaches used, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures.*

Where appropriate, include line diagrams to display the GHG data collection and management system.

Monitoring Plan Summary

Parameter	How it's measured for your Modules	Frequency
Biomass Source	Origin of waste (Coconut fronds, Cane trash, Paddy straw, waste agri-residues etc.)	Per Batch
Pyrolysis Temp	Monitored via sensors on the module	Continuous
Biochar Carbon Content	Laboratory testing of samples	Per Batch/ Quarterly
End-Use Verification	Receipts/GPS logs of where biochar was applied	Application/Per Sale